MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

**Project** **Mansourah Massarah Gold Project**
**Customer:** **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

**Title:** **DESIGN CALCULATION**
**3420 PRESSURE OXDN.- AUTOCL. BLDG.**
**DESIGN OF FLASH V. SUPPORT FDN.**

**Document Number:** **MD-216-3420-EG-CV-CAL-1002**
**Revision:** **D01**
**Revision Date:** **18.11.2023**
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## DEFINITIONS AND ABBREVIATIONS

| | | |
|---|---|---|
| Document | : | Information and the medium on which it is contained (ISO 9000:2015) |
| EMPLOYER | : | 'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives |
| Contractor | : | Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract). |
| 'PMT' | : | Ma'aden Project Management Team |
| 'MMGP Project' | : | Mansourah Massarah Gold Processing Project |
| Attribute | : | Information element that describes a document or tag. |
| Supplier | : | An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor |
| Sub-Supplier | : | A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor. |
| QA | : | Quality Assurance |
| QC | : | Quality Control |
| Works | : | Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities |
| AUTOCL. | : | Autoclave |
| OXDN. | : | Oxidation |
| FDN. | : | Foundation |
| FLASH V. | : | Flash Vessel |
| BLDG. | : | Building |


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## 1. INTENT

This document is to prepare design of foundation for **Flash vessel (3420-FSH-001) support structure**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

## 2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

## 3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.


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# 4. DESIGN REQUIREMENTS

## 4.1 Design Load

All Design Loads are indicated in Load Data drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

# 5. CONCRETE FOUNDATIONS

## 5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

## 5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)


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## 5.3 Minimum Concrete Cover

| Concrete Exposure | Member | Minimum Cover(mm) |
|---|---|---|
| Exposed to weather or contact with ground | Foundation | 75 |
| Exposed to weather or contact with ground | Column | 50 |

## 5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 475 Kn/$m^2$ at 3.0 m below EGL. ( Refer Table 5 - BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).


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## 6. DESIGN CALCULATION FOR FOUNDATION


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**Load Calculation**

Unit weight of soil = 18 kN/m3
Height of backfill soil = 2.3 m
Load on Foundation = 41.4 kN/m2
= 4.14 Ton/m2
Surcharge load on foundation = 20 kN/m2
= 2 Ton/m2

**Stability Check**

Allowable Bearing Capacity of Soil, $Q_{all}$ = 475 kN/m$^2$

| Load Case | Loads at bottom of foundation level V (kN) | Qmax= V/A + M/Z kN/m2 | Soil Bearing Check |
|---|---|---|---|
| LC201 | 7284.00 | 223.00 | OK |
| LC202 | 4385.00 | 119.00 | OK |
| LC203 | 9969.00 | 319.00 | OK |
| LC204 | 4978.00 | 141.00 | OK |
| LC205 | 7056.00 | 230.00 | OK |
| LC206 | 3224.00 | 93.00 | OK |

**Sliding Check**

Calculate safety factor against sliding by dividing resisting shear by actual shear
Fs =H resistance force / H acting force > 1.5

| Load Case | Loads at bottom of V (kN) | Coef. of friction | Hresist (kN) | Hacting (kN) | Fs | Sliding Check |
|---|---|---|---|---|---|---|
| LC201 | 7284.00 | 0.3 | 2185.20 | 0.00 | - | OK |
| LC202 | 4385.00 | 0.3 | 1315.50 | 0.00 | - | OK |
| LC203 | 9969.00 | 0.3 | 2990.70 | 0.00 | - | OK |
| LC204 | 4978.00 | 0.3 | 1493.40 | 0.00 | - | OK |
| LC205 | 7056.00 | 0.3 | 2116.80 | 0.00 | - | OK |
| LC206 | 3224.00 | 0.3 | 967.20 | 0.00 | - | OK |

**Overturning Check**

Calculate safety factor against overturning by dividing resisting moment by overturning moment
The resultant Safety Factor should be 1.5 minimum
Fs =M resistance moment / M overturning moment > 1.5

| Load Case | Loads at bottom of V (kN) | Edge Dis.(d/2) | Mresist (kNm) | Macting (kNm) | Fs | Overturning Check |
|---|---|---|---|---|---|---|
| LC201 | 7284.00 | 3.2 | 23308.80 | 0.00 | - | OK |
| LC202 | 4385.00 | 3.2 | 14032.00 | 0.00 | - | OK |
| LC203 | 9969.00 | 3.2 | 31900.80 | 0.00 | - | OK |
| LC204 | 4978.00 | 3.2 | 15929.60 | 0.00 | - | OK |
| LC205 | 7056.00 | 3.2 | 22579.20 | 0.00 | - | OK |
| LC206 | 3224.00 | 3.2 | 10316.80 | 0.00 | - | OK |


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Characteristic strength of concrete, f 'c= 35 MPa
Yield stress of steel reinforcement, fy= 420 MPa
Clear cover for main reinforcement bars = 50mm

**Raft Design**

Raft Thickness = 700 mm
STAAD output:- ( Refer Attachment 1)

Max BM (+) (My)= 192.344 kN m/m (MOMENT IN BOTTOM FACE)
Max BM (-) (My)= 370.775 kN m/m ( MOMENT IN TOP FACE)
Max BM (+) (Mx)= 191.024 kN m/m (MOMENT IN BOTTOM FACE)
Max BM (-) (Mx)= 370.961 kN m/m ( MOMENT IN TOP FACE)

Max Shear Qx = 0.40 N/mm2
Max Shear Qy = 0.40 N/mm2

Effective depth of Raft in shorter direction = 615.00 mm
Effective depth of Raft in longer direction = 595.00 mm

| Moment | Mux (kN-m) | $\phi$ Kn = $M_{u1}/bd^2$ ($N/mm^2$) | Kn ($N/mm^2$) | $\rho$= | Ast ($mm^2$) | Minm. Ast ($mm^2$) | Bar Dia | Spacing reqd.(mm) | Maxm spacing (mm) | Spacing provd. (mm) | Ast ($mm^2$) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| My(+) | 192.34 | 0.51 | 0.68 | 0.002 | 1004.45 | 1107.00 | D16 | 181.63 | 300 | 150 | 1340.41 |
| My(-) | 370.78 | 0.98 | 1.31 | 0.003 | 1957.92 | 1107.00 | D20 | 160.46 | 300 | 150 | 2094.40 |
| Mx(+) | 191.02 | 0.54 | 0.72 | 0.002 | 1031.83 | 1071.00 | D16 | 187.73 | 300 | 150 | 1340.41 |
| Mx(-) | 370.96 | 1.05 | 1.40 | 0.003 | 2028.04 | 1071.00 | D20 | 154.91 | 300 | 150 | 2094.40 |

Check against one-way shear force

Allowable shear stress, ($\phi$ Vc)/bd = 0.74 $N/mm^2$
Shear stress at a distance 'd' from raft top , $t_v$= 0.40 $N/mm^2$ **OK**

Summary of reinforcement in raft **700 mm**
(HORIZONTAL REIN. IN BOTTOM FACE) **D16 150 c/c**
(HORIZONTAL REIN. IN TOP FACE) **D20 150 c/c**
(HORIZONTAL REIN. IN BOTTOM FACE) **D16 150 c/c**
(HORIZONTAL REIN. IN TOP FACE) **D20 150 c/c**


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## REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.


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# ATTACHMENT 1 PLATE CENTER STRESS SUMMARY


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# Plate Center Stress Summary

| | | | Shear | | Membrane | | | Bending | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Plate | L/C | Qx ($kN/m^2$) | Qy ($kN/m^2$) | Sx ($kN/m^2$) | Sy ($kN/m^2$) | Sxy ($kN/m^2$) | Mx (kNm/m) | My (kNm/m) | Mxy (kNm/m) |
| Max Qx | 112 | 103:1.2DL-1+1 | **1.43E+3** | 1.44E+3 | 0.000 | 0.000 | 0.000 | 137.560 | 152.969 | -61.456 |
| Min Qx | 97 | 103:1.2DL-1+1 | **-1.41E+3** | 1.38E+3 | 0.000 | 0.000 | 0.000 | 137.740 | 148.246 | 46.838 |
| Max Qy | 112 | 103:1.2DL-1+1 | 1.43E+3 | **1.44E+3** | 0.000 | 0.000 | 0.000 | 137.560 | 152.969 | -61.456 |
| Min Qy | 31 | 103:1.2DL-1+1 | 1.38E+3 | **-1.4E+3** | 0.000 | 0.000 | 0.000 | 144.079 | 133.673 | 50.965 |
| Max Sx | 9 | 101:1.4DL-1 | -860.983 | -860.902 | **0.000** | 0.000 | 0.000 | 79.669 | 79.740 | -39.304 |
| Min Sx | 9 | 101:1.4DL-1 | -860.983 | -860.902 | **0.000** | 0.000 | 0.000 | 79.669 | 79.740 | -39.304 |
| Max Sy | 9 | 101:1.4DL-1 | -860.983 | -860.902 | 0.000 | **0.000** | 0.000 | 79.669 | 79.740 | -39.304 |
| Min Sy | 9 | 101:1.4DL-1 | -860.983 | -860.902 | 0.000 | **0.000** | 0.000 | 79.669 | 79.740 | -39.304 |
| Max Sxy | 9 | 101:1.4DL-1 | -860.983 | -860.902 | 0.000 | 0.000 | **0.000** | 79.669 | 79.740 | -39.304 |
| Min Sxy | 9 | 101:1.4DL-1 | -860.983 | -860.902 | 0.000 | 0.000 | **0.000** | 79.669 | 79.740 | -39.304 |
| Max Mx | 152 | 103:1.2DL-1+1 | -1.04E+3 | -1.04E+3 | 0.000 | 0.000 | 0.000 | **191.024** | 184.192 | -45.512 |
| Min Mx | 157 | 103:1.2DL-1+1 | -3.463 | -31.543 | 0.000 | 0.000 | 0.000 | **-370.961** | -3.288 | -2.075 |
| Max My | 140 | 103:1.2DL-1+1 | 1.03E+3 | 1.03E+3 | 0.000 | 0.000 | 0.000 | 185.828 | **192.344** | -43.692 |
| Min My | 193 | 103:1.2DL-1+1 | 31.565 | 21.872 | 0.000 | 0.000 | 0.000 | -3.275 | **-370.775** | -3.046 |
| Max Mxy | 38 | 103:1.2DL-1+1 | 371.047 | -391.757 | 0.000 | 0.000 | 0.000 | -115.977 | -129.616 | **90.206** |
| Min Mxy | 92 | 103:1.2DL-1+1 | 396.294 | 391.507 | 0.000 | 0.000 | 0.000 | -118.489 | -113.691 | **-97.735** |

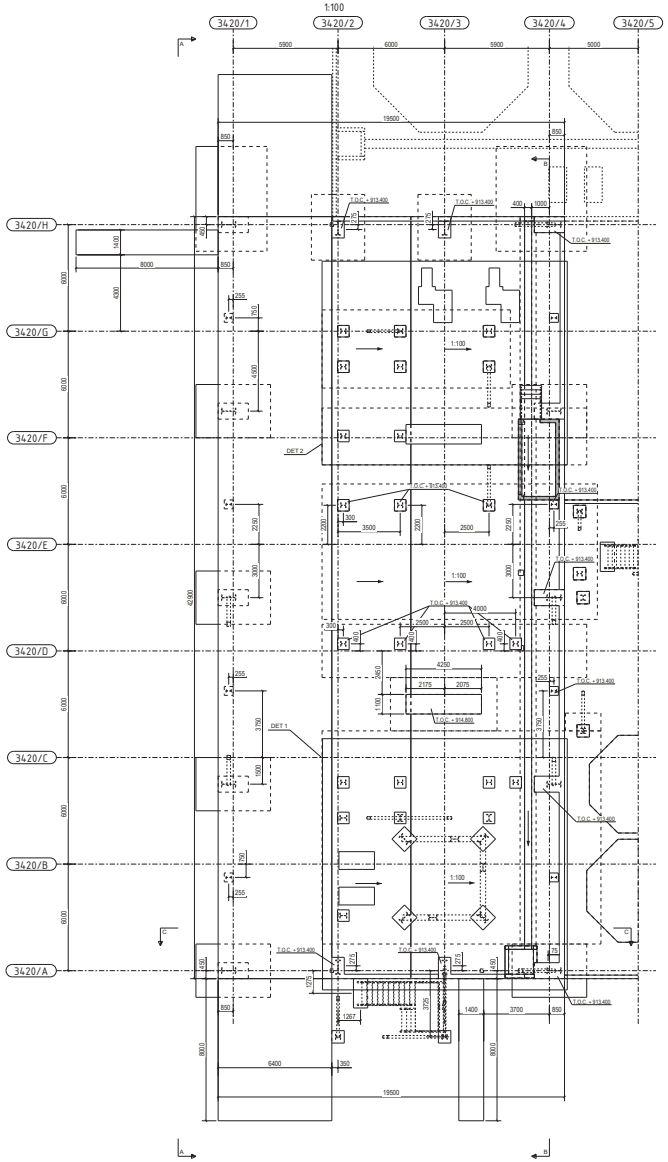
## ATTACHMENT 2 LOAD DATA DRAWINGS


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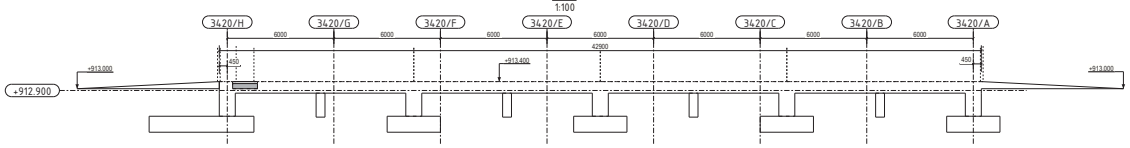
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MD-216-3420-EG-CV-GAD-1001
A01

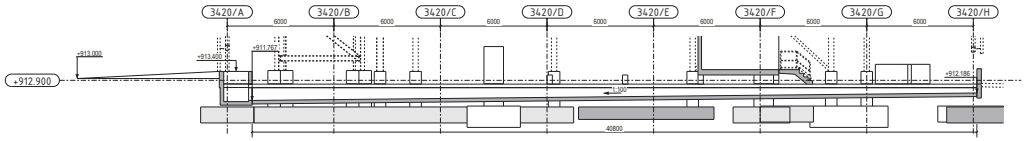
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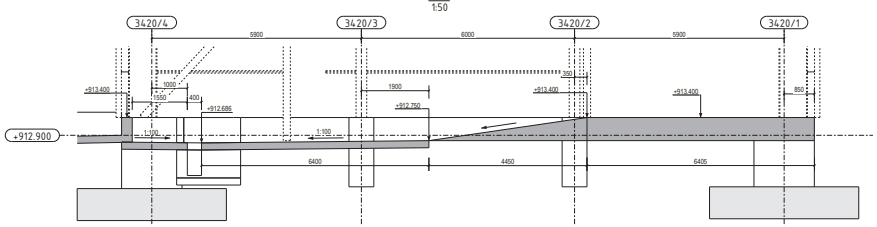

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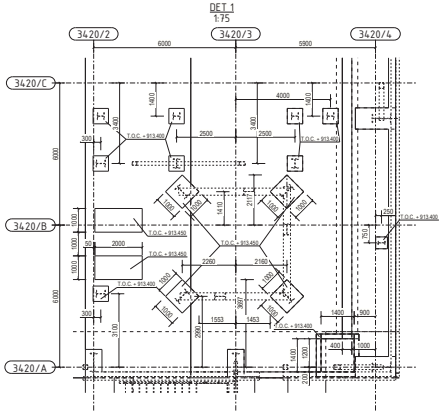

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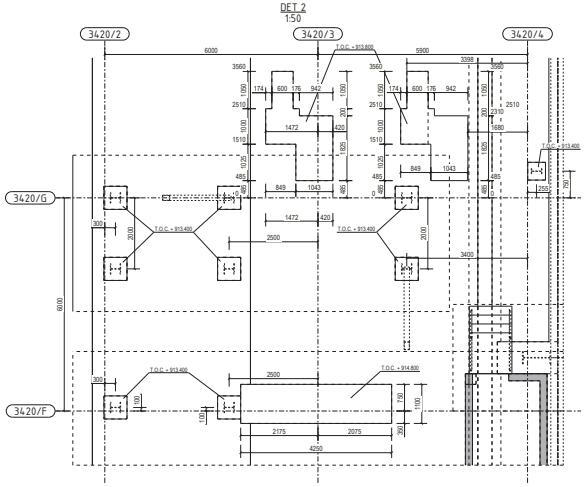

C – C
1:50




DET 1
1:75




DET 2
1:50



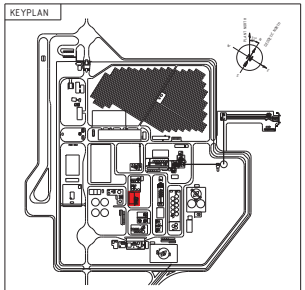

THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3420 = RL(+)912.900 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



| DOCUMENT NUMBER | REFERENCE DOCUMENTS | DOCUMENT NUMBER | REFERENCE DOCUMENTS |
|---|---|---|---|
| MD-216-3420-EG-CV-GAD-1002 | PLAN AT EL. +912.900 | MD-216-3420-EG-ST-GAD-1001 | PLAN AT EL. +913.400 & +914.570 |
| MD-216-3420-EG-ME-GAD-1003 | SECTION C-C | MD-216-3420-EG-ME-GAD-1004 | SECTION F-F |
| MD-216-3420-EG-ME-GAD-1005 | PLAN AT EL. +913.400 TO EL. +930.950 | MD-216-3420-EG-ME-GAD-1006 | SECTION D-D |
| MD-216-9000-EG-ST-CRT-0001 | CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS | MD-216-3420-EG-ME-GAD-1006 | PLAN AT EL. +913.400 |

| REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | CMP |
|---|---|---|---|---|---|---|---|
| A01 | 20.12.2019 | ISSUED FOR APPROVAL | Melan | MALai | LuiPei | SiouOw | |

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company

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| LARSEN & TOUBRO DRAWING NUMBER | REVISION |
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| OUTOTEC DRAWING NUMBER | DOCUMENT ID / REVISION |
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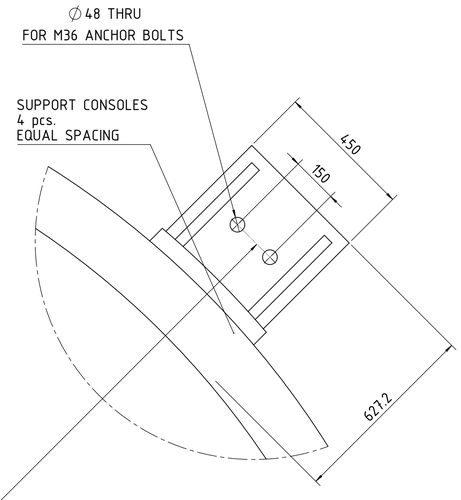
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| DRAWN | Melan | 20.12.2019 |
| CHECKED | MALai | 20.12.2019 |
| ENGINEER | LuiPei | 20.12.2019 |
| APPROVED | SiouOw | 20.12.2019 |
| EMPLOYER | | |
| SCALE | 1:100 | A0 |

MANSOURAH – MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
PRESSURE OXIDATION – AUTOCL. FLASH V. & SCRUBBER
PLAN AT EL. +913.400

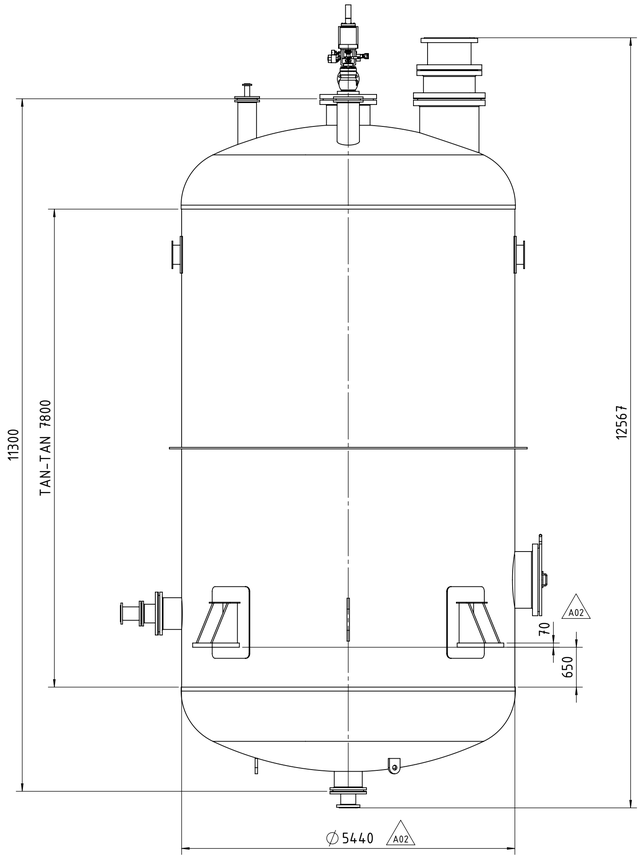
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REV: A01

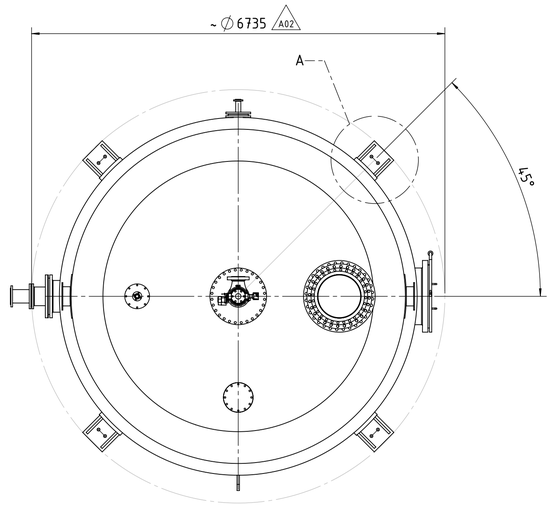
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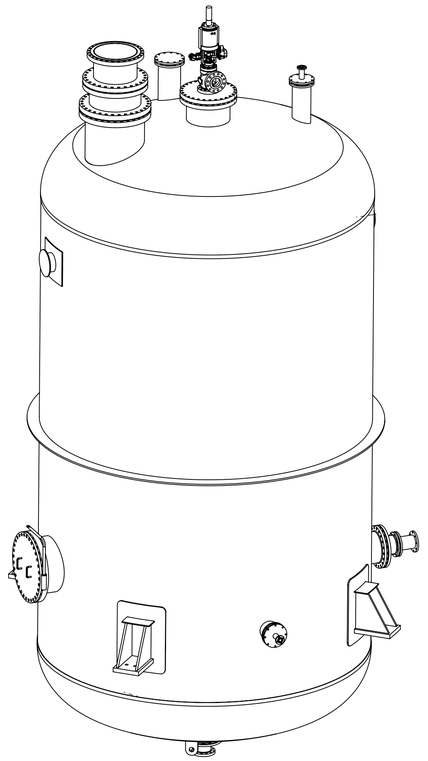
PLANT NORTH




DETAIL A
SCALE 1 : 10



GENERAL TOLERANCES
ISO 2768-1 EN ISO 13920 EN ISO 9013 DIN 6935
Values in millimetres

| | Basic size range | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LINEAR DIMENSIONS | > 0,5 ± 3 | > 3 ± 6 | > 6 ± 30 | > 30 ± 120 | > 120 ± 400 | > 400 ± 1000 | > 1000 ± 2000 | > 2000 ± 4000 | > 4000 ± 8000 | > 8000 ± 12 000 | > 12 000 ± 16 000 | > 16 000 ± 20 000 | > 20 000 |
| Machining ISO 2768-m Chamfers and external radii | ± 0,1 ± 0,2 | ± 0,1 ± 0,5 | ± 0,2 ± 1 | ± 0,3 | ± 0,5 | ± 0,8 | ± 1,2 | ± 2 | | | | | |
| Welding ISO 13920-B | ± 1 (Range 2–30) | | | ± 2 | ± 2 | ± 3 | ± 4 | ± 6 | ± 8 | ± 10 | ± 12 | ± 14 | ± 16 |
| Thermal cutting ISO 9013-2 3,15 < t ≤ 6,3 6,3 < t ≤ 10 10 < t ≤ 50 | ≥ 3 < 10 ± 0,7 ± 1 ± 1,8 | ≥ 10 < 35 ± 0,8 ± 1,1 ± 1,8 | ≥ 35 < 125 ± 0,9 ± 1,3 ± 1,8 | ≥ 125 < 315 ± 1,1 ± 1,4 ± 1,9 | ≥ 315 < 1000 ± 1,2 ± 1,5 ± 2,3 | ≥ 1000 < 2000 ± 1,3 ± 1,6 ± 3 | ≥ 2000 < 4000 ± 1,3 ± 1,7 ± 4,2 | Quality of cut surface ISO 9013-44 | | | | | |
| ANGULAR DIMENSIONS | Shorter side ≤ 10 | > 10 ≤ 50 | > 50 ≤ 120 | > 120 ≤ 400 | > 400 ≤ 1000 | > 1000 | | | | | | | |
| Machining ISO 2768-m | ± 1° | ± 0°30' | ± 0°20' | ± 0°10' | ± 0°5' | | | | | | | | |
| Welding ISO 13920-B | ± 13 mm/m | | | | ± 9 mm/m | ± 6 mm/m | | | | | | | |
| Bending | SEE DIN 6935 | | | | | | | | | | | | |
| GEOMETRICAL TOLERANCES | Machining ISO 2768-K Welding ISO 13920-F | | | | | | | | | | | | |

GENERAL GEOMETRICAL TOLERANCES
ISO 2768-2 EN ISO 13920
Values in millimetres

| | Basic size range | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| MACHINING ISO 2768-K | ≤ 100 | > 100 ≤ 300 | > 300 ≤ 1000 | > 1000 ≤ 3000 | ≤ 10 | > 10 ≤ 30 | > 30 ≤ 100 | > 100 ≤ 300 | > 300 ≤ 1000 | > 1000 ≤ 3000 |
| Perpendicularity | 0,4 | 0,6 | 0,8 | 1 | | | | | | |
| Symmetry | 0,6 | 0,6 | 0,8 | 1 | | | | | | |
| Straightness and flatness | | | | | 0,05 | 0,1 | 0,2 | 0,4 | 0,6 | 0,8 |
| Circular run-out | 0,2 | | | | | | | | | |

| | Basic size range | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| WELDING ISO 13920-F | ≤ 30 ± 120 | > 120 ± 400 | > 400 ± 1000 | > 1000 ± 2000 | > 2000 ± 4000 | > 4000 ± 8000 | > 8000 ± 12 000 | > 12 000 ± 16 000 | > 16 000 ± 20 000 | > 20 000 |
| Straightness, flatness and parallelism | 1 | 1,5 | 3 | 4,5 | 6 | 8 | 10 | 12 | 14 | 16 |

GRAPHICAL SYMBOLS FOR WELDING ARE IN ACCORDANCE WITH ISO 2553:2014

WELDED JOINTS WITHOUT SPECIFICATION OF QUALITY CLASS ACCORDING TO ISO 5817-C

GEOMETRIC TOLERANCES PER ISO 1101

SURFACE TEXTURES PER ISO 1302

MEASUREMENTS WITHOUT INDICATION OF TOLERANCES

| MACHINING | THERMAL CUTTING | WELDMENT |
|---|---|---|
| ISO 2768-m | ISO 9013 | ISO 13920-B |

CORNERS PER ISO 13715 (-0.3 / +0.3)

NOTES

Vessel is located indoors. No wind load considered.

Reported seismic load is a primary load, derived in accordance with the site specifications and the requirements of SBC301 and is suitable for combination in accordance with SBC301 for either strength design OR allowable stress design.
Combination factor for strength design shall be 1.0,
however if the loads are to be used for allowable stress design method then the combination factor would be 0.7.

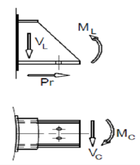
Seismic Conditions:

1° Code: IBC 2018, Hazardous Content: YES, Site Class: D, Importance Factor: 1.5, Ss = 0.05g, S1 = 0.03g, Risk Category: II ---> Seismic Desing Category: A
2° Code: IBC 2018, Hazardous Content: YES, Site Class: D, Importance Factor: 1.5, Ss = 0.10g, S1 = 0.02g, Risk Category: II ---> Seismic Desing Category: A
} Give Same Coefficient to PGA 0.02g

Wind Conditions: No Wind Considered. Vessel is Indoor.

**BASE LOADS**

| LOAD CASE | ±Pr (kN) | VL (kN) | ML (kNm) | Vc (kN) | Mc (kNm) |
|---|---|---|---|---|---|
| Weight, Operating CORRODED | 0 | 948 | 0 | 0 | 0 |
| Weight, Operating NEW | 0 | 948 | 0 | 0 | 0 |
| Weight, Empty NEW | 0 | 209 | 0 | 0 | 0 |
| Weight, Vacuum CORRODED | 0 | 948 | 0 | 0 | 0 |
| Seismic, Operating NEW | 9 | 978 | 0 | 0 | 0 |
| Seismic, Empty NEW | 2 | 216 | 0 | 0 | 0 |
| Seismic, Vacuum CORRODED | 9 | 978 | 0 | 0 | 0 |
| Wind Case | N/A | N/A | N/A | N/A | N/A |




| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
| MD-216-3420-EG-ME-DDE-3422 | GENERAL ARRANGEMENT DRAWING | A02 | 18.3.2020 | REVISED AS MARKED | JK | PI | MH | GR | |
| MD-216-3420-EG-ME-DAT-3422 | DATASHEET | A01 | 04.02.2020 | BASE LOADS UPDATED; ISSUED FOR APPROVAL | JK | PI | MH | GR | |


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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: O19045_450_M_D_02696_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707_GAU01_HH12_MLD15_00002 — DOCUMENT ID: OU601274888 — REVISION: 2

| | | |
|---|---|---|
| DRAWN | maihan | 18.03.2020 |
| CHECKED | Peklmm | 18.03.2020 |
| ENGINEER | maihan | 18.03.2020 |
| APPROVED | georie | 18.03.2020 |
| EMPLOYER | | |
| SCALE | 1:50 | A1 |

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
PRESSURE OXIDATION
3420-FSH-001 FLASH VESSEL

DRAWING NUMBER: MD-216-3420-EG-ME-PLN-1002 — REV: A02

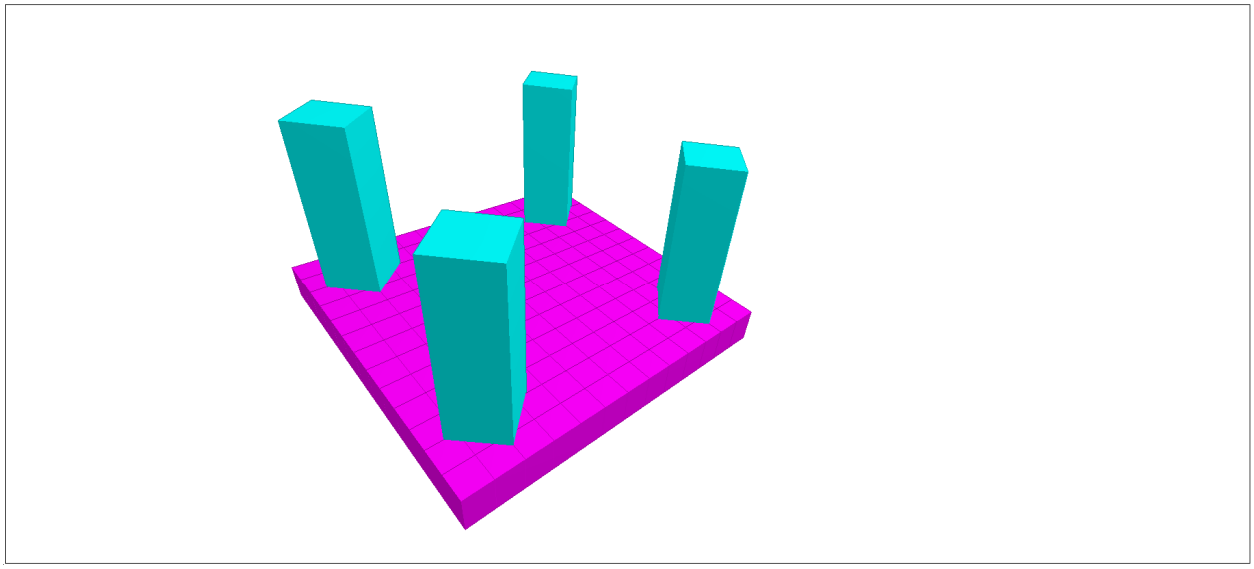
## ATTACHMENT 3 ISOMETRIC VIEW OF STAAD MODEL


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# STAAD.Pro Report

| | | | |
|---|---|---|---|
| To: | From: | | |
| Copy to: | Date: 24/04/2020 15:39:00 | Ref: | ca/ Document1 |



*3D Rendered View*

## ATTACHMENT 4 STAAD INPUT


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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13.04.2020
JOB NAME MMGP/FLASH VESSEL SUPPORT FDN.
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
CHECKER NAME SJM
APPROVED NAME AG
END JOB INFORMATION
UNIT METER MTON
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171 7947 7911 7912 7948; 172 7948 7912 7913 7949;
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175 7952 7938 7914 7950; 176 7894 7895 7953 7954;
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179 7897 7898 7957 7956; 180 7898 7899 7958 7957;
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185 7915 7894 7954 7963; 186 7903 7927 7964 7962;
187 7951 7915 7963 7965; 188 7917 7966 7967 7916;
189 7916 7967 7968 7918; 190 7918 7968 7969 7919;
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193 7921 7971 7972 7922; 194 7922 7972 7973 7923;
195 7923 7973 7974 7924; 196 7924 7974 7975 7925;
197 7925 7975 7976 7926; 198 7926 7976 7977 7927;
199 7939 7978 7966 7917; 200 7927 7977 7979 7964;
ELEMENT PROPERTY
9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
200 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.75
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40262
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 1 ZD 1
CONSTANTS
BETA 45 MEMB 2
BETA 135 MEMB 1
BETA 315 MEMB 3
BETA 225 MEMB 4
MATERIAL CONCRETE ALL
SUPPORTS
9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
200 PLATE MAT DIRECT Y SUBGRADE 5700
*************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD _ OPERATING NEW  ( DL -1  )
SELFWEIGHT Y -1
JOINT LOAD
7777 FX 0.02 FY -95.614 FZ 0.02
7779 FX 0.02 FY -95.614 FZ -0.02
7781 FX -0.02 FY -95.614 FZ -0.02
7783 FX -0.02 FY -95.614 FZ 0.02
ELEMENT LOAD
* LOAD DUE TO BACKFILL
9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
200 PR GY -4.15
* LOAD DUE TO SURCHARGE
9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 200 PR GY -2
*************************************************************************************************
```





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*************************************************************************************************
LOAD 2 LOADTYPE None  TITLE DEAD LOAD _ EMPTY  ( DL -2  )
SELFWEIGHT Y -1
JOINT LOAD
7777 FX 0.02 FY -21.714 FZ 0.02
7779 FX 0.02 FY -21.714 FZ -0.02
7781 FX -0.02 FY -21.714 FZ -0.02
7783 FX -0.02 FY -21.714 FZ 0.02
ELEMENT LOAD
* LOAD DUE TO BACKFILL
9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
200 PR GY -4.15
* LOAD DUE TO SURCHARGE
9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 200 PR GY -2
*************************************************************************************************
LOAD 3 LOADTYPE None  TITLE SEISMIC LOAD _ OPERATING NEW  ( E -1  )
JOINT LOAD
7777 FX 0.559 FY -98.93 FZ -0.438
7779 FX -0.342 FY -97.913 FZ 0.342
7781 FX 0.438 FY -98.93 FZ -0.559
7783 FX 0.617 FY -95.427 FZ -0.617
*************************************************************************************************
LOAD 4 LOADTYPE None  TITLE SEISMIC LOAD _ EMPTY  ( E -2  )
JOINT LOAD
7777 FX 0.124 FY -21.851 FZ -0.097
7779 FX -0.076 FY -21.625 FZ 0.076
7781 FX 0.097 FY -21.851 FZ -0.124
7783 FX 0.137 FY -21.073 FZ -0.137
*************************************************************************************************
* FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL-1
1 1.4
LOAD COMB 102 1.4DL-2
2 1.4
LOAD COMB 103 1.2DL-1+1.0E-1
1 1.2 3 1.0
LOAD COMB 104 1.2DL-2+1.0E-2
2 1.2 4 1.0
LOAD COMB 105 0.9DL-1+1.0E-1
1 0.9 3 1.0
LOAD COMB 106 0.9DL-2+1.0E-2
2 0.9 4 1.0
*************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
LOAD COMB 201 1.0DL-1
1 1.0
LOAD COMB 202 1.0DL-2
2 1.0
LOAD COMB 203 1.0DL-1+0.7E-1
1 1.0 3 0.7
LOAD COMB 204 1.0DL-2+0.7E-2
2 1.0 4 0.7
LOAD COMB 205 0.6DL-1+0.7E-1
1 0.6 3 0.7
LOAD COMB 206 0.6DL-2+0.7E-2
2 0.6 4 0.7
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 3500 ALL
CLB 0.05 ALL
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 20 ALL
MINMAIN 20 ALL
MINSEC 10 ALL
DESIGN COLUMN 1 TO 4
END CONCRETE DESIGN
FINISH
```

## ATTACHMENT 5 STAAD OUTPUT


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```
          **************************************************
          *                                                *
          *          STAAD.Pro V8i SELECTseries5           *
          *          Version  20.07.10.64                  *
          *          Proprietary Program of                *
          *          Bentley Systems, Inc.                 *
          *          Date=    APR 22, 2020                 *
          *          Time=    13:25:15                     *
          *                                                *
          *     USER ID: LNT ECC DIV                       *
          **************************************************
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    1. STAAD SPACE
INPUT FILE: FLASH VESSEL SUPPORT - FDN..STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 13.04.2020
    4. JOB NAME MMGP/FLASH VESSEL SUPPORT FDN.
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. CHECKER NAME SJM
   10. APPROVED NAME AG
   11. END JOB INFORMATION
   12. UNIT METER MTON
   13. JOINT COORDINATES
   14. 7777 369.842 913.45 -383.293; 7779 369.842 913.45 -387.747
   15. 7781 365.388 913.45 -387.747; 7783 365.388 913.45 -383.293
   16. 7784 369.842 909.9 -383.293; 7785 369.842 909.9 -387.747
   17. 7786 365.388 909.9 -387.747; 7787 365.388 909.9 -383.293
   18. 7788 365.388 909.9 -387.247; 7789 369.842 909.9 -387.247
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114. MEMBER INCIDENCES
115. 1 7784 7777; 2 7785 7779; 3 7786 7781; 4 7787 7783
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189. ELEMENT PROPERTY
190. 9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
191. 200 THICKNESS 0.7
192. DEFINE MATERIAL START
193. ISOTROPIC STEEL
194. E 2.09042E+007
195. POISSON 0.3
196. DENSITY 7.75
197. ALPHA 1.2E-005
198. DAMP 0.03
199. TYPE STEEL
200. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
201. ISOTROPIC CONCRETE
202. E 2.21467E+006
203. POISSON 0.17
204. DENSITY 2.40262
205. ALPHA 1E-005
206. DAMP 0.05
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207. TYPE CONCRETE
208. STRENGTH FCU 2812.28
209. END DEFINE MATERIAL
210. MEMBER PROPERTY AMERICAN
211. 1 TO 4 PRIS YD 1 ZD 1
212. CONSTANTS
213. BETA 45 MEMB 2
214. BETA 135 MEMB 1
215. BETA 315 MEMB 3
216. BETA 225 MEMB 4
217. MATERIAL CONCRETE ALL
218. SUPPORTS
219. 9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
220. 200 PLATE MAT DIRECT Y SUBGRADE 5700
221. ******************************************************************************
222. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD _ OPERATING NEW  ( DL -1  )
223. SELFWEIGHT Y -1
224. JOINT LOAD
225. 7777 FX 0.02 FY -95.614 FZ 0.02
226. 7779 FX 0.02 FY -95.614 FZ -0.02
227. 7781 FX -0.02 FY -95.614 FZ -0.02
228. 7783 FX -0.02 FY -95.614 FZ 0.02
229. ELEMENT LOAD
230. * LOAD DUE TO BACKFILL
231. 9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
232. 200 PR GY -4.15
233. * LOAD DUE TO SURCHARGE
234. 9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 200 PR GY -2
235. ******************************************************************************
236. LOAD 2 LOADTYPE NONE  TITLE DEAD LOAD _ EMPTY  ( DL -2  )
237. SELFWEIGHT Y -1
238. JOINT LOAD
239. 7777 FX 0.02 FY -21.714 FZ 0.02
240. 7779 FX 0.02 FY -21.714 FZ -0.02
241. 7781 FX -0.02 FY -21.714 FZ -0.02
242. 7783 FX -0.02 FY -21.714 FZ 0.02
243. ELEMENT LOAD
244. * LOAD DUE TO BACKFILL
245. 9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 199 -
246. 200 PR GY -4.15
247. * LOAD DUE TO SURCHARGE
248. 9 12 17 18 21 24 29 TO 94 97 99 101 103 105 107 109 111 TO 200 PR GY -2
249. ******************************************************************************
250. LOAD 3 LOADTYPE NONE  TITLE SEISMIC LOAD _ OPERATING NEW  ( E -1  )
251. JOINT LOAD
252. 7777 FX 0.559 FY -98.93 FZ -0.438
253. 7779 FX -0.342 FY -97.913 FZ 0.342
254. 7781 FX 0.438 FY -98.93 FZ -0.559
255. 7783 FX 0.617 FY -95.427 FZ -0.617
256. ******************************************************************************
257. LOAD 4 LOADTYPE NONE  TITLE SEISMIC LOAD _ EMPTY  ( E -2  )
258. JOINT LOAD
259. 7777 FX 0.124 FY -21.851 FZ -0.097
260. 7779 FX -0.076 FY -21.625 FZ 0.076
261. 7781 FX 0.097 FY -21.851 FZ -0.124
262. 7783 FX 0.137 FY -21.073 FZ -0.137
```





```
263. ******************************************************************************
264. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
265. LOAD COMB 101 1.4DL-1
266. 1 1.4
267. LOAD COMB 102 1.4DL-2
268. 2 1.4
269. LOAD COMB 103 1.2DL-1+1.0E-1
270. 1 1.2 3 1.0
271. LOAD COMB 104 1.2DL-2+1.0E-2
272. 2 1.2 4 1.0
273. LOAD COMB 105 0.9DL-1+1.0E-1
274. 1 0.9 3 1.0
275. LOAD COMB 106 0.9DL-2+1.0E-2
276. 2 0.9 4 1.0
277. ******************************************************************************
278. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
279. LOAD COMB 201 1.0DL-1
280. 1 1.0
281. LOAD COMB 202 1.0DL-2
282. 2 1.0
283. LOAD COMB 203 1.0DL-1+0.7E-1
284. 1 1.0 3 0.7
285. LOAD COMB 204 1.0DL-2+0.7E-2
286. 2 1.0 4 0.7
287. LOAD COMB 205 0.6DL-1+0.7E-1
288. 1 0.6 3 0.7
289. LOAD COMB 206 0.6DL-2+0.7E-2
290. 2 0.6 4 0.7
291. ******************************************************************************
292. PERFORM ANALYSIS
```

**P R O B L E M  S T A T I S T I C S**

**NUMBER OF JOINTS 200 NUMBER OF MEMBERS 4**
**NUMBER OF PLATES 169 NUMBER OF SOLIDS 0**
**NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 196**

**SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER**

**ORIGINAL/FINAL BAND-WIDTH= 143/ 24/ 75 DOF**
**TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 612**
**TOTAL LOAD COMBINATION CASES = 12 SO FAR.**
**SIZE OF STIFFNESS MATRIX = 46 DOUBLE KILO-WORDS**
**REQRD/AVAIL. DISK SPACE = 13.4/ 147299.5 MB**

```
293. LOAD LIST 101 TO 106
294. START CONCRETE DESIGN
```

```
295. CODE ACI
296. TRACK 2 ALL
297. FC 3500 ALL
298. CLB 0.05 ALL
299. CLS 0.05 ALL
300. CLT 0.05 ALL
301. MAXMAIN 20 ALL
302. MINMAIN 20 ALL
303. MINSEC 10 ALL
304. DESIGN COLUMN 1 TO 4
```





```
================================================================

     COLUMN  NO.     1  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  34.3 MPA,  SQRE SIZE -1000.0 X1000.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED = 10000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
----------------------------------------------------------

 32 - 20 MM               1.005      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ----------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
33040.32  26432.26  13111.79   4813.64     367.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 1875.15  -4158.84   2019.56      0.00    0.00000
 ----------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ----------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
33040.32  26432.26  13111.79   4813.64     367.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 1875.15  -4158.84   2019.56      0.00    0.00000
 ----------------------------------------------------------
                        Pn       Mn       Pn       Mn    (@ Z )
          |            24399.01  3247.57 12199.50  4776.94
       P0 |*           22365.76  3723.98 10166.25  4610.65
          | *          20332.51  4107.33  8133.00  4319.15
    Pn,max|__*         18299.26  4400.85  6099.75  3902.39
          |   *        16266.00  4611.40  4066.50  3359.02
    Pn    |    *       14232.75  4756.16  2033.25  2678.47
   NOMINAL|     *         Pn       Mn       Pn       Mn    (@ Y )
     AXIAL|      *     24399.01  3247.57 12199.50  4776.94
COMPRESSION|       *    22365.76  3723.98 10166.25  4610.65
        Pb|-------*Mb  20332.51  4107.33  8133.00  4319.15
          |      *     18299.26  4400.85  6099.75  3902.39
__________|____*_______16266.00  4611.40  4066.50  3359.02
          |  * M0   Mn,14232.75  4756.16  2033.25  2678.47
          | *   BENDING
    P-tens|*     MOMENT
          |
```

**COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 34.3 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 32 - 20 MM | 1.005 | 101 | END | 0.650 |

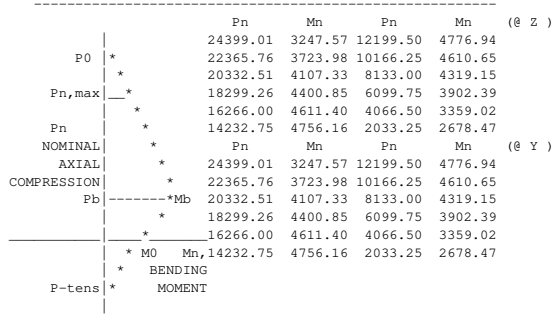
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 33040.32 | 26432.26 | 13111.79 | 4813.64 | 367.1 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 1875.15 | -4158.84 | 2019.56 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 33040.32 | 26432.26 | 13111.79 | 4813.64 | 367.1 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 1875.15 | -4158.84 | 2019.56 | 0.00 | 0.00000 |








**COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 34.3 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 32 - 20 MM | 1.005 | 101 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 33040.32 | 26432.26 | 13111.79 | 4813.64 | 367.1 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 1875.15 | -4158.84 | 2019.56 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 33040.32 | 26432.26 | 13111.79 | 4813.64 | 367.1 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 1875.15 | -4158.84 | 2019.56 | 0.00 | 0.00000 |

==========

**COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 34.3 MPA, SQRE SIZE -1000.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 10000.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 32 - 20 MM | 1.005 | 101 | END | 0.650 |

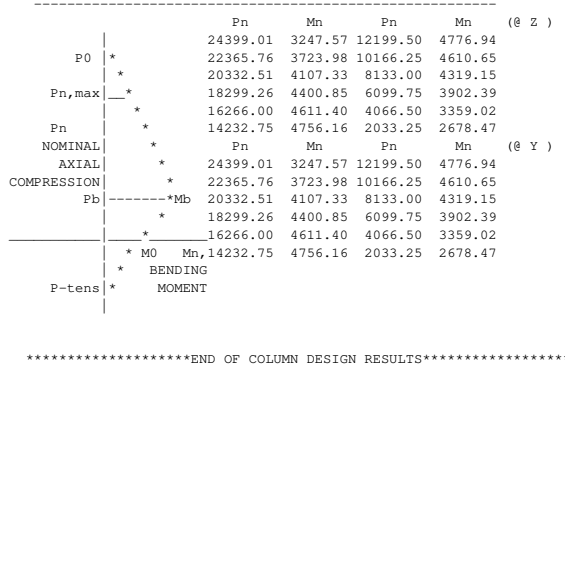
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 33040.32 | 26432.26 | 13111.79 | 4813.64 | 367.1 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 1875.15 | -4158.84 | 2019.56 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 33040.32 | 26432.26 | 13111.79 | 4813.64 | 367.1 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 1875.15 | -4158.84 | 2019.56 | 0.00 | 0.00000 |




**********END OF COLUMN DESIGN RESULTS**********





305. END CONCRETE DESIGN
306. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 22,2020 TIME= 13:25:20 ****

**********
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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